MIL-V-46704A(AT)
10 September 1968
SUPERSEDING
MIL-V-46704(Ord)
1 March 1962

MILITARY SPECIFICATION

VALVE, RELAY EMERGENCY: BREAKAWAY AIR BRAKE

1. SCOPE

1.1 This specification covers the minimum requirements for an air operated relay emergency valve used in vehicle air brake systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-704	- Treatment and Painting of Materiel.
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL-F-13927	- Fungus Resistance Test; Automotive Components.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of U S Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MS33586	- Metals, Definition of Dissimilar.
MS53004	- Valve, Relay, Emergency-Breakaway, Air Brake.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

FSC 2530

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3.1 First article. Unless otherwise specified, (see 6.2) the contractor shall furnish the first article samples consisting of four valves, for first article inspection to determine conformance to 4.2. The samples shall be fully representative of the valves to be supplied from production facilities and tooling.

3.2 Materials. Materials and processes used in the manufacture of these valves shall conform to applicable Government specifications. Materials conforming to supplier's specifications may be used, provided the specifications are satisfactory to the Government and contain provisions for adequate tests. The use of supplier's specification will not constitute waiver of Government inspection.

3.2.1 Metals. All metals used in the construction of these valves shall be of a corrosion-resisting type or shall be adequately protected to resist corrosion during the normal service life of the valve and shall meet the requirements of 3.5.5. The use of dissimilar metals which would encourage galvanic action, especially brass, copper, or steel, in metal to metal contact with aluminum or aluminum alloy, shall be avoided wherever practicable. Dissimilar metals are defined in MS33586.

3.3 Design and construction. The valve shall be designed and constructed to the form and dimensions specified herein and on MS53004.

3.3.1 Locking devices. Lockwashers, self-locking nuts, or other approved locking devices shall be incorporated where specified or where required to prevent loosening of components.

3.3.2 Port marking. All ports shall be clearly and permanently marked and identified to indicate the proper air line application and connections to be made. Decalcomanias shall not be considered permanent marking.

3.4 Performance requirements.

3.4.1 Cracking pressure. When tested as specified in 4.5.1, the service line pressure required to actuate the valve allowing air pressure to enter delivery line shall be not more than 5.0 pounds per square inch (psi).

3.4.2 Charging rate. When tested as specified in 4.5.2, the time required to build up trailer reservoir pressure from 0 to 65 psi shall be 7 $\pm$ 1 second, and from 0 to 100 psi shall be not more than 30 seconds.

3.4.3. Pressure differential. When tested as specified in 4.5.3, the pressure differential between the service line pressure and the brake chamber pressure shall be not more than 5.0 psi.

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3.4.4 Service time. When tested as specified in 4.5.4, the time between application of 100 psi air pressure to the service port and build up of 60 psi in the brake chamber shall be a maximum of 0.22 seconds.

3.4.5 Emergency application. When tested as specified in 4.5.5, the pressure shall increase in the 75 cubic inch reservoir at approximately three times the rate that the emergency line pressure decreases. The time between the start of emergency line pressure decrease and the build up of 60 psi in the 75 cubic inch reservoir shall not exceed .19 seconds.

3.4.6 Bleed-down rate. When tested as specified in 4.5.6, the rate at shich the air vents through the bleed-off shall be as specified on MS53004. Air shall not bleed back through the supply line.

3.4.7 Leakage. When tested as specified in 4.5.7, the soap bubble leakage, with reservoir at 100 psi and brake chamber at atmospheric air pressure, shall not exceed a 1 inch diameter bubble in 5 seconds. When under normal service application, the bubble leakage shall not exceed a 1 inch diameter bubble in 2 seconds.

3.4.8 Endurance.

3.4.8.1 Service applications. When tested as specified in 4.5.8.1, the valve shall have a life expectancy of 1,000,000 cycles of operation. A cycle of operation shall consist of applying 50 $\pm$ 5 psi to the service port, then releasing service port pressure until brake chamber returns to atmospheric pressure.

3.4.8.2 Emergency applications. When tested as specified in 4.5.8.2, the valve shall have a life expectancy of 50,000 cycles of operation. A cycle of operation shall consist of dropping the emergency line pressure below 45 psi allowing pressure build up in the brake chamber to 50 $\pm$ 5 psi, then applying pressure to the emergency line above 45 psi until brake chamber is at atmospheric air pressure.

3.5 Environmental requirements.

3.5.1 Extreme temperature.

3.5.1.1 Low temperature. When tested as specified in 4.6.1.1, the valve shall meet the requirements of 3.4.1 through 3.4.6.

3.5.1.2 High temperature. When tested as specified in 4.6.1.2, the valve shall meet the requirements of 3.4.1 through 3.4.6.

3.5.2 Waterproofness. When tested as specified in 4.6.2, the valve shall evidence no leakage and shall meet the requirements of 3.4.1 through 3.4.6.

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3.5.3 Shock resistance. When tested as specified in 4.6.3, components of the valve shall not be loosened or damaged, nor shall the valve malfunction when so tested. After testing, the valve shall meet the requirements of 3.4.1 through 3.4.6.

3.5.4 Vibration resistance. When tested as specified in 4.6.4, components of the valve shall not be loosened or damaged, nor shall the valve malfunction when so tested. After testing, the valve shall meet the requirements of 3.4.1 through 3.4.6.

3.5.5 Corrosion resistance. When tested as specified in 4.6.5, the valve shall evidence no corrosion affecting operation and shall meet the requirements of 3.4.1 through 3.4.6.

3.5.6 Sand and dust resistance. When tested as specified in 4.6.6, the valve shall evidence no damage affecting operation and shall meet the requirements of 3.4.1 through 3.4.6.

3.5.7 Humidity resistance. When tested as specified in 4.6.7, the valve shall evidence no damage affecting operation and shall meet the requirements of 3.4.1 through 3.4.6.

3.5.8 Fungus resistance. When tested as specified in 4.6.8, the valve shall evidence no fungus growth affecting operation and shall meet the requirements of 3.4.1 through 3.4.6.

3.6 Finish. External surfaces of the valve shall be treated and painted in accordance with the applicable provisions as specified in MIL-T-704.

3.7 Marking. The valve shall be marked as specified on MS53004.

3.8 Workmanship. Workmanship shall be of a quality which will assure a product free of burrs, rust, scratches, chips, sharp edges, or other defects which will affect serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection. First article inspection shall be conducted by the Government at a place designated by the procuring activity (see 6.2), and shall consist of the examination specified in 4.3.2.2 and testing in the order specified in table I. Failure of the first article samples to pass any examination or test specified herein shall be cause, at the option of the Government, for refusal to conduct additional testing until the faults revealed by the tests have been corrected.

Table I. First article testing.

Sample	Paragraph	Test
1, 2, 3, and 4	4.5.1	Cracking pressure
	4.5.2	Charging rate
	4.5.3	Pressure differential
	4.5.4	Service time
	4.5.5	Emergency application
	4.5.6	Bleed-down rate
	4.5.7	Leakage
1	4.6.7	Humidity resistance
	4.5.8	Endurance
2	4.6.3	Shock resistance
	4.6.4	Vibration resistance
	4.6.8	Fungus resistance
3	4.6.1	Extreme temperature
	4.6.5	Corrosion resistance
4	4.6.2	Waterproofness
	4.6.6	Sand and dust resistance

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 Lot formation. Unless otherwise specified (see 6.2), a lot shall consist of all valves from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with inspection level II of MIL-STD-105.

4.3.1.3 Sampling for testing. Samples for quality conformance testing shall be selected in accordance with inspection level S3 of MIL-STD-105.

4.3.2 Quality conformance examination.

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4.3.2.1 Acceptable quality level. Each sample selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL), on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.3.2.2 Classification of defects. For examination purposes, defects shall be classified as specified in table II. Any valve in the sample containing one or more defects shall be rejected and if the number of defective valves in any sample exceeds the quality conformance number for that sample, the lot represented by the sample shall be rejected.

Table II. Classification of defects.

Categories	Defects	Method of inspection
Critical:	None defined	
Major:		
101	Dimensions affecting interchangeability not within tolerance (see 3.3)	Gage
Minor:		
201	Dimensions not affecting interchangeability not within tolerance (see 3.3)	Gage
202	Improper marking (see 3.7)	Visual
203	Faulty workmanship (see 3.8)	Visual

4.3.3 Testing.

4.3.3.1 Quality conformance tests. Samples selected in accordance with 4.3.1.3 shall be subjected to the tests specified in table III, in the order specified, using an AQL of 6.5 on the basis of percent defective.

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Table III. Quality conformance testing.

Paragraph	Test
4.5.1	Cracking pressure
4.5.2	Charging rate
4.5.3	Pressure differential
4.5.4	Service time
4.5.5	Emergency application
4.5.6	Bleed-down rate
4.5.7	Leakage

4.4 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the valves shall be examined to determine compliance with the requirements of section 5 of this specification.

4.5 Performance tests.

4.5.1 Cracking pressure. After mounting on a test stand simulating the actual conditions of operation, the valve shall be actuated by allowing air pressure to build up through the service line until diaphragm or relay piston operates. A manometer or other accurate means of measuring pressure shall be used to determine pressure required to actuate valve. The test shall be repeated no less than five times and the results recorded to determine conformance to 3.4.1.

4.5.2 Charging rate. Air pressure shall be allowed to flow through the emergency port, passing through the check valve and to a reservoir of 1,000 cubic inches. Time required to build up pressure from 0 to 65 psi in the reservoir shall be recorded. Repeat test no less than five times. Then record the time required to build up pressure from 0 psi to 100 psi. Repeat test no less than five times. All results shall be recorded to determine conformance to 3.4.2.

4.5.3 Pressure differential. An air pressure of 50 $\pm$ 5 psi shall be applied to the service port. The air pressure on the service piston or diaphragm shall be greater than the air pressure in the brake chamber at all times. Pressure differential shall be observed and recorded. The test shall consist of no less than five service applications with results recorded to determine conformance to 3.4.3.

4.5.4 Service time. An air pressure of 100 psi shall be applied to the service port and the time required to build up to 60 psi in the brake chamber recorded. The test shall consist of no less than five service applications with results recorded to determine conformance to 3.4.4.

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4.5.5 Emergency application. The emergency line shall be gradually vented to atmospheric pressure permitting pressure from the trailer reservoir to build up in the brake chamber. Determine the ratio of the metered controlling pressure between the emergency line pressure drop and the brake chamber pressure build up. After recharging the emergency line (releasing brake chamber pressure), vent the emergency line to atmospheric pressure rapidly as in trailer breakaway. Determine the time required to build up 60 psi in the brake chamber. The test shall consist of no less than five emergency application cycles with results recorded to determine conformance to 3.4.5.

4.5.6 Bleed-down rate. With all other ports closed, a constant air pressure within the limits specified on MS 53004 shall be applied to the delivery port of the valve. Measure the bleed-down rate at which the air vents through the bleed-off. Pressure must be maintained for no less than 3 minutes. The results shall be recorded to determine conformance to 3.4.6.

4.5.7 Leakage. With reservoir charged to 100 psi and brake chamber at atmospheric pressure, the exhaust port shall be coated with soap suds then tested for leakage. With a minimum of 60 psi in the brake chamber, the exhaust port shall be coated again with soap suds and tested for leakage. The emergency line pressure shall be reduced below 45 psi by putting the valve into emergency. The exhaust port shall be coated with soap suds and tested for leakage. In each of the leakage test phases, the time required to create a 1 inch diameter bubble to determine conformance to 3.4.7 shall be recorded.

4.5.8 Endurance.

4.5.8.1 Service applications. The valve shall be subjected to an atmospheric temperature of minus 65° ± 5°F or a period of 4 hours. While stabilized at this temperature, with the reservoir charged to 100 psi and the brake chamber at atmospheric pressure, the valve shall be subjected to a duty cycle of five service applications (see 3.4.8.1). Cycling shall be at 30 cycles per minute (cpm). Allow the valve to warm to room temperature (77° ± 15°F) and continue life cycling to a total of 1,000,000 service applications. After completion of test, continue with emergency applications.

4.5.8.2 Emergency applications. After the above test, the valve shall again be subjected to atmospheric temperature of minus 65° ± 5°F for a period of 4 hours. Then, with the reservoir charged to 100 psi and the brake chamber at atmospheric pressure, the valve shall be subjected to a duty cycle of five emergency applications (see 3.4.8.2). Cycling shall be at 30 cpm. Allow the valve to warm to room temperature and continue life cycling to a total of 50,000 emergency applications. After completion of service and emergency cycling tests, conduct visual inspection for internal and external structural failures. The valve shall then be subjected to the tests specified in 4.5.1 through 4.5.6 to determine conformance to 3.4.8.

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4.6 Environmental tests.

4.6.1 Extreme temperature.

4.6.1.1 Low temperature. The valve shall be placed in a test chamber and cooled to and maintained at a temperature of minus 80° ± 5°F for a period of 48 hours. The temperature of the chamber shall then be raised to, and the valve stabilized at, minus 65° ± 5°F. After being stabilized at that temperature, the valve shall be subjected to the test specified in 4.5.1 through 4.5.6 to determine conformance to 3.5.1.1.

4.6.1.2 High temperature. The valve shall be placed in a test chamber and heated to and maintained at a temperature of plus 165° ± 5°F for a period of 120 hours. A relative humidity of not more than 15 percent shall be maintained in the test chamber throughout the exposure period. The temperature of the chamber shall then be lowered to and the valve stabilized at plus 125° ± 5°F. After being stabilized at that temperature, the valve shall be subjected to the tests specified in 4.5.1 through 4.5.6 to determine conformance to 3.5.1.2.

4.6.2 Waterproofness. The valve shall be subjected to the test for waterproofness in accordance with method 104A of MIL-STD-202. Failure of the valve to pass any of the tests specified therein shall constitute nonconformance to 3.5.2.

4.6.3 Shock resistance. The valve shall be subjected to the shock test as specified in procedure V of MIL-E-5272. After the test, the valve shall be subjected to the tests specified in 4.5.1 through 4.5.6 to determine conformance to 3.5.3.

4.6.4 Vibration resistance. The valve shall be rigidly mounted on a vibrating machine and subjected to the vibration test specified in method 201 of MIL-STD-202 for 1 hour in the direction of each of the three major axes. After the test, the valve shall be subjected to the tests specified in 4.5.1 through 4.5.6 to determine conformance to 3.5.4.

4.6.5 Corrosion resistance. The valve shall be subjected to the salt spray (fog) test as specified in method 101 of MIL-STD-202 using a solution of 20 percent, except that the duration of the test shall be 200 hours. After the test, the valve shall be subjected to the tests specified in 4.5.1 through 4.5.6 to determine conformance to 3.5.5.

4.6.6 Sand and dust resistance. The valve shall be subjected to procedure I, sand and dust resistance test of MIL-E-5272, except that the test shall be conducted at an atmospheric temperature of plus 125° ± 5°F only and the test period shall be 24 hours. The valve shall be mounted in the dust chamber as in intended use. After the test, the valve shall be subjected to the tests specified in 4.5.1 through 4.5.6 to determine conformance to 3.5.6.

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4.6.7 Humidity resistance. The valve shall be placed in the test chamber and the temperature stabilized at minus 40° ± 5°F. After a period of 2 hours, the temperature shall be raised to plus 140° ± 5°F. Stabilize and maintain for 2 hours. The temperature shall then be reduced to 77° ± 15°F. Humidity need not be controlled during the pre-test phase. The temperature shall be raised gradually to plus 160° ± 5°F and the test continued as specified in procedure I of MIL-E-5272 with the relative humidity maintained at 95 percent for a total of 10 cycles. After the test, the valve shall be subjected to the tests specified in 4.5.1 through 4.5.6 to determine conformance to 3.5.7.

4.6.8 Fungus resistance. The valve shall be subjected to the class 1, method B fungus resistance test of MIL-F-13927, except that the test period shall be a continuous 90 days. After the test, the valve shall be subjected to the tests specified in 4.5.1 through 4.5.6 to determine conformance to 3.5.8.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, packing, and marking shall be in accordance with the applicable packaging standard or packaging data sheet specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The relay emergency valve covered by this specification is intended for use in air operated braking systems of military vehicles.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article if required (see 3.1).
(c) Military part number (see 3.7).
(d) Place where first article shall be sent (see 4.2).
(e) Any special lot size (see 4.3.1.1)
(f) Selection of applicable packaging standard or packaging data sheet (see 5.1).

Custodian:
Army - AT

Preparing activity:
Army - AT

Review activity:
Army - MI

Project No. 2530-A101

User activity:
Army - ME

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-V-46704A(AT)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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